LoRA: Low-Rank Adaptation of Large Language Models

Background

Motivation

- Full fine-tuning is expensive and scales poorly
- Question: Can LoRA achieve comparable performance on GLUE?

LoRA (Hu et al., 2022)

- Freeze backbone model, inject **low-rank matrices** (A, B)

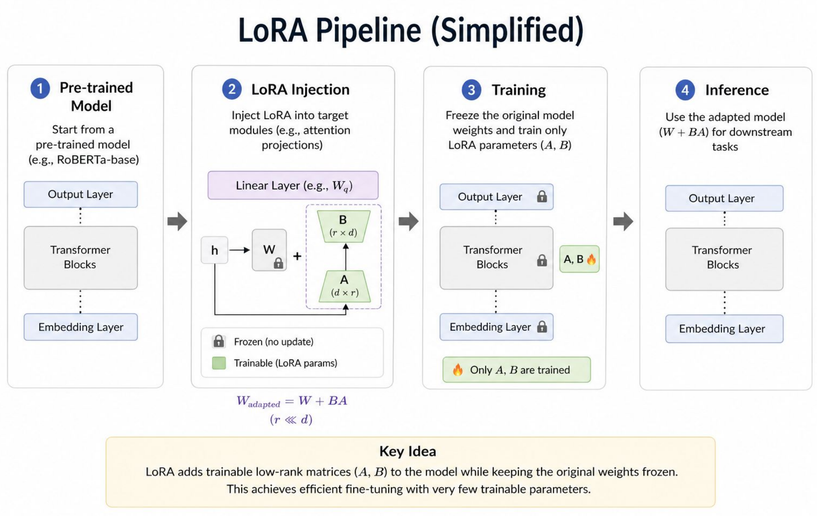
Project Goal

- Re-implement LoRA on RoBERT-base (Liu et al., 2019)
- LoRA vs Full fine-tuning vs Frozen backbone
- The effect of LoRA rank, alpha and target modules

Project Setup 🤗

- Model: RoBERTa-base
- Dataset: GLUE benchmark tasks (MNLI, MRPC, QNLI, QQP, SST-2, CoLA)
- Target Modules: Query, Value
- Compute Resource: NVIDIA A800, PyTorch+HuggingFace
- Batch size=32, Training epochs=3, Max Sequence Length=128

Methodology



Future Work

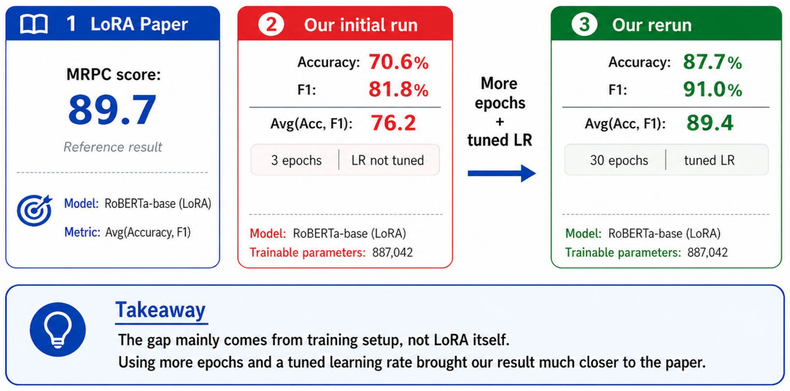
- Expand the efficiency analysis by measuring GPU memory usage and storage cost in addition to trainable parameters and runtime.
- Extend the experiments to larger backbone models to test whether the same trends hold beyond RoBERTa-base.
- Run more multi-seed experiments to improve result stability, especially on smaller and more variable tasks.

Experiment Results

Method	MNLI	MRPC	QNLI	QQP	SST-2	CoLA
Frozen Backbone	53.34	68.38	66.30	75.14	80.62	69.13
Full Fine-tuning	**85.98**	**87.25**	**91.09**	80.41	91.97	**82.64**
LoRA	84.95	70.59	90.48	**87.36**	**93.46**	79.39

Table 1: Performance comparison of Frozen Backbone, Full Fine-tuning, and LoRA on GLUE tasks.

Why was our initial MRPC result much lower than the LoRA paper?



Takeaway

The gap mainly comes from training setup, not LoRA itself.
Using more epochs and a tuned learning rate brought our result much closer to the paper.

Extension

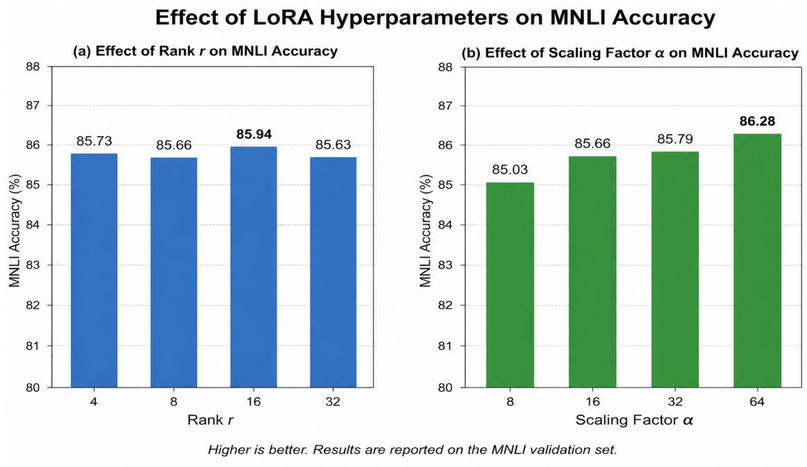


Table 7: MNLI performance comparison for different LoRA target modules.

LoRA Target Modules	Eval Loss	Accuracy	Precision	Recall	F1
query	0.4694	81.43%	81.31%	81.32%	81.31%
query, value	0.3932	84.92%	84.80%	84.83%	84.81%
query, value, key, dense	0.3716	85.92%	85.82%	85.84%	85.83%

Conclusion

Overall Performance on GLUE benchmark

- LoRA achieves performance close to, and sometimes better than full fine-tuning on multiple tasks (QQP, SST-2).

Peformance Efficiency

- LoRA uses only about 0.89M trainable parameters (0.71%), a roughly 140x reduction vs full fine-tuning.

Effect of hyperparameters

- Accuracy improves steadily as alpha increases from 8 to 64
- Increasing rank does not guarantee monotonic improvement
- Best weight modules in practice: query+value.

References

- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., & Chen, W. (2022). LoRA: Low-Rank Adaptation of Large Language Models. ICLR.
- Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., & Bowman, S. (2018). GLUE: A Multi-Task Benchmark and Analysis Platform for Natural Language Understanding.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach.

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